

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$\frac{7}{2}x + 6y = 25$$
$$\frac{5}{2}x + 6y = 23$$

The solution to the given system of equations is (x, y) . What is the value of $\frac{17}{2}x + 18y$?

Answer

- A. 2
- B. 3
- C. 48
- D. 71

Correct Answer: D

Rationale

Choice D is correct. Multiplying the second equation in the given system by 2 yields $\frac{10}{2}x + 12y = 46$. Adding this equation to the first equation in the system yields $(\frac{7}{2}x + 6y) + (\frac{10}{2}x + 12y) = 25 + 46$, which is equivalent to $(\frac{7}{2}x + \frac{10}{2}x) + (6y + 12y) = 25 + 46$, or $\frac{17}{2}x + 18y = 71$. Therefore, the value of $\frac{17}{2}x + 18y$ is 71.

Choice A is incorrect. This is the value of x , not the value of $\frac{17}{2}x + 18y$.

Choice B is incorrect. This is the value of y , not the value of $\frac{17}{2}x + 18y$.

Choice C is incorrect. This the value of $(\frac{7}{2}x + 6y) + (\frac{5}{2}x + 6y)$, or $6x + 12y$, not the value of $\frac{17}{2}x + 18y$.